Expansion of the Coriolis Term in the N-S Equations

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The Coriolis Term

in rotating reference frame N-S Equations

The **Navier-Stokes Equations** in a rotating reference frame:

$$\frac{\partial \mathbf{U}}{\partial t} + (\mathbf{U} \cdot \nabla)\mathbf{U} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{U} \underline{- 2\mathbf{\Omega} \times \mathbf{U}} - \mathbf{g} \tag{1}$$

- The Coriolis acceleration: $-2\mathbf{\Omega} \times \mathbf{U}$.
- In component form (for geophysical flows):

$$\begin{cases} \frac{\mathrm{D}u_1}{\mathrm{D}t} \underline{- f u_2} = -\frac{1}{\rho}\frac{\partial p}{\partial x_1} + \nu \nabla^2 u_1 \\ \frac{\mathrm{D}u_2}{\mathrm{D}t} \underline{+ f u_1} = -\frac{1}{\rho}\frac{\partial p}{\partial x_2} + \nu \nabla^2 u_2 \end{cases} \tag{2}$$

- where the `Coriolis parameter` f is defined as: $f = 2\Omega \sin\phi$.

Then how to **EXPAND** the Coriolis term from Eq. (1) to Eq. (2) ?

Cross Product: $\mathbf{A} \times \mathbf{B}$

using Tensor Notation

We define a **permutation symbol**, ε_{ijk} as:

Levi-Civita symbol ε_{ijk}

$$\mathbf{e}_i \times \mathbf{e}_j = \varepsilon_{ijk}\mathbf{e}_k, \quad \varepsilon_{ijk} = \begin{cases} 1 & @ \quad i,j,k = 123123...\text{(even)} \\ -1 & @ \quad i,j,k = 321321...\text{(odd)} \\ 0 & @ \quad i,j,k: \ \text{at least one repeated} \end{cases}$$

Therefore we can define a cross product:

$$\mathbf{A} \times \mathbf{B} = (a_i\mathbf{e}_i) \times (b_j\mathbf{e}_j) = a_i b_j \varepsilon_{ijk}\mathbf{e}_k = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

where $\mathbf{e}_i$ and $\mathbf{e}_j$ denote the unit basis vectors.

Decomposition of the Angular Velocity Vector

To analyze the Coriolis effect at a specific latitude ϕ, we decompose the Earth's angular velocity vector $\boldsymbol{\Omega}$ into the local Cartesian coordinate system $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$.

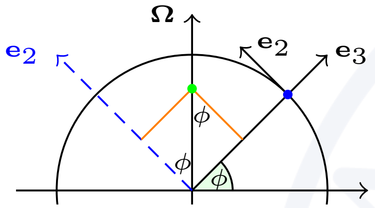

Fig 1. Coordinate transformation between the global rotating frame and the local Cartesian frame $(\mathbf{e}_2, \mathbf{e}_3)$. And the $\mathbf{e}_1$ component is omitted as it is orthogonal to the Earth's axis. So we ignore the $\mathbf{e}_1$.

`Vector Decomposition`: In the local frame $\boldsymbol{\Omega}$ is expressed as:

$$\boldsymbol{\Omega} = \Omega \cos\phi \mathbf{e}_2 + \Omega \sin\phi \mathbf{e}_3$$

Defining the velocity vector $\mathbf{U}$ in index notation:

$$\mathbf{U} = u_j \mathbf{e}_j = u_1\mathbf{e}_1 + u_2\mathbf{e}_2 + u_3\mathbf{e}_3$$

The Coriolis force:

$$-2\boldsymbol{\Omega} \times \mathbf{U} = -2\Omega\cos\phi \mathbf{e}_2 \times u_j\mathbf{e}_j - 2\Omega\sin\phi\mathbf{e}_3 \times u_j\mathbf{e}_j$$

Derivation

Using the permutation symbol:

$$
\begin{aligned}
-2\boldsymbol{\Omega} \times \mathbf{U} &= -2\Omega \cos\phi \mathbf{e}_2 \times u_j \mathbf{e}_j - 2\Omega \sin\phi \mathbf{e}_3 \times u_j \mathbf{e}_j \\
&= -2\Omega \cos\phi u_j \varepsilon_{2jk} \mathbf{e}_k - 2\Omega \sin\phi u_j \varepsilon_{3jk} \mathbf{e}_k \\
&= -2\Omega[\cos\phi(u_1\varepsilon_{213}\mathbf{e}_3 + u_3\varepsilon_{231}\mathbf{e}_1) + \sin\phi(u_1\varepsilon_{312}\mathbf{e}_2 + u_2\varepsilon_{321}\mathbf{e}_1)] \\
&= -2\Omega[\cos\phi(-u_1\mathbf{e}_3 + u_3\mathbf{e}_1) + \sin\phi(u_1\mathbf{e}_2 - u_2\mathbf{e}_1)] \\
&= -2\Omega[(\cos\phi u_3 - \sin\phi u_2)\mathbf{e}_1 + \sin\phi u_1 \mathbf{e}_2 - \cos\phi u_1 \mathbf{e}_3]
\end{aligned}
$$

Hydrostatic Approximations: the vertical momentum equation is dominated by the Hydrostatic Balance, and neglect the u_3. Then

$$
\begin{aligned}
-2\boldsymbol{\Omega} \times \mathbf{U} &= 2\Omega \sin\phi u_2 \mathbf{e}_1 - 2\Omega \sin\phi u_1 \mathbf{e}_2 \\
&= f u_2 \mathbf{e}_1 - f u_1 \mathbf{e}_2
\end{aligned}
$$

where the $f = 2\Omega \sin\phi$.

Alternatively

If you can calculate determinants:

$$
\begin{aligned}
-2\boldsymbol{\Omega} \times \mathbf{U} &= -2 \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ 0 & \Omega\cos\phi & \Omega\sin\phi \\ u_1 & u_2 & u3 \end{vmatrix} \\
&= -2\Omega[(-u_2\sin\phi + u_3\cos\phi)\,\mathbf{e}_1 + u_1\sin\phi\mathbf{e}_2 - u_1\cos\phi\mathbf{e}_3] \\
&= fu_2\mathbf{e}_1 - fu_1\mathbf{e}_2
\end{aligned}
$$

where the $f = 2\Omega\sin\phi$. So

$$
\begin{cases}
\dfrac{\partial u_1}{\partial t} + u_1\dfrac{\partial u_1}{\partial x_1} + u_2\dfrac{\partial u_1}{\partial x_2} + u_3\dfrac{\partial u_1}{\partial x_3} = -\dfrac{1}{\rho}\dfrac{\partial p}{\partial x_1} + \nu\nabla^2 u_1 + fu_2 \\
\dfrac{\partial u_2}{\partial t} + u_1\dfrac{\partial u_2}{\partial x_1} + u_2\dfrac{\partial u_2}{\partial x_2} + u_3\dfrac{\partial u_2}{\partial x_3} = -\dfrac{1}{\rho}\dfrac{\partial p}{\partial x_2} + \nu\nabla^2 u_2 - fu_1
\end{cases}
$$

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